

Lelin Zheng (She/Her)

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EDUCATION

Northeastern University

M.S. in Computer Science, GPA: 4.0 / 4.0

Coursework:

Object-Oriented Design, Algorithms, Database Management Systems, Cloud Computing, Distributed Systems, DevOps

Sep 2024 – May 2027

Seattle, WA

TECHNICAL KNOWLEDGE

- **Languages:** Python, Java, JavaScript, HTML/CSS, C, TypeScript, SQL
- **Tools & Frameworks:** Spring Boot 3, Spring Cloud (Gateway, Eureka, Config), Resilience4j, Hibernate (JPA), Angular, React, Node.js, Express.js, Sails.js, Bootstrap, Tailwind CSS, Flask, Django, PyQt6, Docker, Kubernetes, Helm, RabbitMQ, Linux/Unix, Nginx, Git
- **Testing & Observability:** JUnit, Jest, Supertest, WireMock, Grafana Stack (Prometheus, Loki, Tempo)
- **Databases & Services:** PostgreSQL, MySQL, SQLite, MongoDB Atlas, AWS (EC2, S3, IAM, CloudFront, SQS)

RELEVANT WORK EXPERIENCE

Software Engineer Intern (Incoming) – Block Inc.

- Incoming intern focused on backend systems and scalable services

May 2026 – Aug 2026

Seattle, WA

Software Engineer Intern

Jetsweat Inc.

- Built **10+ RESTful APIs** to surface user-level analytics and power a white-label admin toggle, enabling **50+ corporate partners** to gain insights while reducing **response latency by 40%** through optimized queries and connection pooling.
- Redesigned MySQL schema by adding an access-control column and enforcing role-based SQL queries to ensure **100% restriction** of unauthorized admin access to individual video engagement data, aligning with security best practices.
- Revamped the onboarding modal with **React**, using `useState` for step control and `Redux` to persist authenticated user context across views, improving recommendation personalization and driving a **25% increase in content engagement**.
- Enhanced the admin dashboard (JavaScript, Node.js, EJS, Bootstrap) with real-time engagement tracking and error surfacing, resolving **10+ production issues** and increasing active dashboard usage by **30%**.

May 2025 – Aug 2025

New York, NY

High School Computer Science Teacher

Calgary Board of Education, Crescent Heights High School

- Taught Computer Science to **100+ students** in grades 10 and 11, covering programming fundamentals, algorithms, procedural and functional programming, OOP in Python, and HTML/CSS for web development.
- Developed a Python-based scheduling application that efficiently organized a 10-team inter-school badminton tournament, reducing manual coordination time by over 80%.

Sep 2022 – Jun 2024

Calgary, AB

PROJECTS

Distributed CI/CD Orchestration Platform — Spring Boot + Kubernetes

- Architected a distributed CI/CD platform across three **Spring Boot microservices** (API Gateway, Execution, Report) with **Resilience4j circuit breaking**, idempotent job handling, and a fail-open DB write strategy to isolate failures under load.
- Decoupled pipeline submission from execution via **RabbitMQ**; the execution worker clones Git repos via JGit, resolves job ordering through topological sort, and runs jobs in isolated Docker containers with fail-fast and per-job overrides.
- Containerized services with Docker and deployed to Kubernetes via Helm with HPA for independent scaling; instrumented Prometheus + Grafana for observability and validated read-path performance under **100 concurrent** users with **~365 req/s** throughput, p95 latency **<12ms**, and 0% error rate.

ChatFlow: A Distributed WebSocket Chat System

- Built a **scalable WebSocket chat system** (Spring Boot, AWS ALB) with **RabbitMQ topic exchange** and 20+ room queues; implemented **AMQP channel pooling**, circuit breakers, and **Redis Pub/Sub fanout** with message deduplication.
- Designed an **async write-behind pipeline** with a dedicated consumer service using manual ACKs, exponential-backoff retries, and DLQ, plus batched PostgreSQL upserts for high-throughput, fault-tolerant persistence.
- Deployed a **7-node AWS system** (EC2, Redis, RabbitMQ, PostgreSQL) with REST analytics APIs and performance monitoring; sustained **5.2K msg/s**, p99 **70ms** latency, and **99.999%** delivery over **5M** messages under concurrent load

Qualcomm On-Device AI Hackathon: AI-Powered Narrative Connect Four (2nd Place Winner)

- Engineered a **PyQt6 GUI** with multi-threaded event handling for a responsive and interactive **LLM-integrated** Connect Four game experience, featuring dynamic UI updates and real-time **AI narration**.
- Integrated **local LLM inference via Ollama (Mistral-7B)** with game logic algorithms (minimax with alpha-beta pruning), implemented speech-to-text feature using **OpenAI Whisper** increasing player-AI interaction by **75%**.